

Validation screenshot

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[(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % minikube start]
minikube v1.38.1 on Darwin 26.5.2 (arm64)
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.50 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
Verifying Kubernetes components...
  Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: default-storageclass, metrics-server, storage-provisioner
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline %
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[(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % kubectl apply -f k8s/training-job.yaml]
job.batch/model-training created
[(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % minikube addons enable metrics-server]
metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
The 'metrics-server' addon is enabled
[(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % kubectl rollout restart deploy/model-serving -n ml-training]
deployment.apps/model-serving restarted
[(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % kubectl get pods -n ml-training -w]
NAME                                READY   STATUS    RESTARTS   AGE
model-serving-5d6757d799-6l9cb      0/1     Running   0           7s
model-serving-b6b78767f-8fgpf       1/1     Running   1 (3d15h ago)  4d7h
model-serving-b6b78767f-f2th9       1/1     Running   1 (3d15h ago)  4d7h
model-training-c76sb                1/1     Running   0           35s
model-serving-5d6757d799-6l9cb      0/1     Running   0           10s
model-serving-5d6757d799-6l9cb      1/1     Running   0           16s
model-serving-b6b78767f-8fgpf       1/1     Terminating 1 (3d15h ago)  4d7h
model-serving-b6b78767f-8fgpf       1/1     Terminating 1 (3d15h ago)  4d7h
model-serving-5d6757d799-d46bn      0/1     Pending    0           0s
model-serving-5d6757d799-d46bn      0/1     Pending    0           0s
model-serving-b6b78767f-8fgpf       0/1     Completed   1 (3d15h ago)  4d7h
model-serving-5d6757d799-d46bn      0/1     Pending    0           0s
model-serving-5d6757d799-d46bn      0/1     ContainerCreating 0           1s
model-serving-b6b78767f-8fgpf       0/1     Completed   1 (3d15h ago)  4d7h
model-serving-b6b78767f-8fgpf       0/1     Completed   1 (3d15h ago)  4d7h
model-serving-5d6757d799-d46bn      0/1     Running     0           2s
model-serving-5d6757d799-d46bn      0/1     Running     0           11s
model-serving-5d6757d799-d46bn      1/1     Running     0           17s
model-serving-b6b78767f-f2th9       1/1     Terminating 1 (3d15h ago)  4d7h
model-serving-b6b78767f-f2th9       1/1     Terminating 1 (3d15h ago)  4d7h
model-serving-b6b78767f-f2th9       0/1     Completed   1 (3d15h ago)  4d7h
model-serving-b6b78767f-f2th9       0/1     Completed   1 (3d15h ago)  4d7h
model-serving-b6b78767f-f2th9       0/1     Completed   1 (3d15h ago)  4d7h
```

```
(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % kubectl get all -n ml-training
kubectl logs -n ml-training job/model-training
kubectl get pods -n ml-training -o wide
kubectl get hpa -n ml-training
```

NAME	READY	STATUS	RESTARTS	AGE
pod/model-serving-59cb68bd56-bfszb	1/1	Running	0	71s
pod/model-serving-59cb68bd56-fplqs	1/1	Running	0	87s
pod/model-training-c76sb	0/1	Completed	0	35m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/model-serving	ClusterIP	10.96.208.247	<none>	80/TCP	4d8h

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/model-serving	2/2	2	2	4d8h

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/model-serving-59cb68bd56	2	2	2	87s
replicaset.apps/model-serving-5bd87fbdd7	0	0	0	4d8h
replicaset.apps/model-serving-5d6757d799	0	0	0	35m
replicaset.apps/model-serving-676cfc9865	0	0	0	4d8h
replicaset.apps/model-serving-b6b78767f	0	0	0	4d7h

NAME	ODS	REPLICAS	AGE	REFERENCE	TARGETS	MINPODS	MAXP
horizontalpodautoscaler.autoscaling/model-serving	2	4d8h		Deployment/model-serving	cpu: <unknown>/70%, memory: 43%/80%	2	5

NAME	STATUS	COMPLETIONS	DURATION	AGE
job.batch/model-training	Complete	1/1	9m8s	35m

```
{
  "event": "env_overrides_applied",
  "overrides": {
    "MAX_EPOCHS": 2,
    "NUM_WORKERS": 2,
    "SUBSET_FRACTION": 0.05
  }
}
{"event": "run_started", "config_path": "/app/configs/training_config.yaml", "device": "cpu", "architecture": "resnet18", "num_classes": 10, "trainable_parameters": 11173962, "epochs": 2, "batch_size": 64, "learning_rate": 0.001, "subset_fraction": 0.05}
{"event": "data_ready", "train_batches": 40, "val_batches": 8, "train_examples": 2500, "val_examples": 500, "data_seconds": 4.5}
{"epoch": 1, "train_loss": 2.0292, "train_accuracy": 0.2632, "val_loss": 1.9178, "val_accuracy": 0.296, "epoch_seconds": 274.8}
{"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt", "epoch": 1}
{"epoch": 2, "train_loss": 1.7498, "train_accuracy": 0.3384, "val_loss": 2.3598, "val_accuracy": 0.256, "epoch_seconds": 260.1}
{"event": "no_improvement", "epoch": 2, "patience_counter": 1}
{"event": "training_complete", "best_val_loss": 1.9178, "best_val_accuracy": 0.296, "checkpoint": "/app/checkpoints/classifier_v1.pt", "training_seconds": 535.1, "data_seconds": 4.5, "total_seconds": 540.0}

```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	READINESS	GATES
model-serving-59cb68bd56-bfszb	1/1	Running	0	71s	10.244.0.20	minikube	<none>	<none>	<none>	<none>
model-serving-59cb68bd56-fplqs	1/1	Running	0	87s	10.244.0.19	minikube	<none>	<none>	<none>	<none>
model-training-c76sb	0/1	Completed	0	35m	10.244.0.16	minikube	<none>	<none>	<none>	<none>

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
model-serving	Deployment/model-serving	cpu: <unknown>/70%, memory: 43%/80%	2	5	2	4d8h

```
(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % curl -s -X POST localhost:8080/predict -F "image=@test_image.png" | python -m json.tool
{
  "filename": "test_image.png",
  "predicted_class": "cat",
  "confidence": 0.2227,
  "probabilities": {
    "airplane": 0.0774,
    "automobile": 0.0124,
    "bird": 0.1779,
    "cat": 0.2227,
    "deer": 0.2091,
    "dog": 0.1492,
    "frog": 0.0588,
    "horse": 0.0076,
    "ship": 0.0724,
    "truck": 0.0125
  }
}
```

```
(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % kubectl get hpa -n ml-training
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
model-serving	Deployment/model-serving	cpu: 0%/70%, memory: 40%/80%	2	5	2	4d8h

```
(.venv) ambethkrishnan@Ambeths-Mac-mini mlops-pytorch-pipeline % curl -s localhost:8080/health && echo && curl -s localhost:8080/metadata | python -m json.tool
{"status": "ok", "model_loaded": true}
{
  "class_names": [
    "airplane",
    "automobile",
    "bird",
    "cat",
    "deer",
    "dog",
    "frog",
    "horse",
    "ship",
    "truck"
  ],
  "architecture": "resnet18",
  "num_classes": 10,
  "checkpoint_path": "/app/checkpoints/classifier_v1.pt",
  "trained_epochs": 1,
  "val_loss": 1.9177873344421388,
  "val_accuracy": 0.296
}
```